AI_TLfanclub at VLSP 2026 ViTNumChart:

Voting over Fine-Tuned Small Multimodal Program Writers, with an Analysis of Single-Writer Errors

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1st of 9
teams on the private leaderboard

0.7917
Program Accuracy, private test
runner-up 0.7866

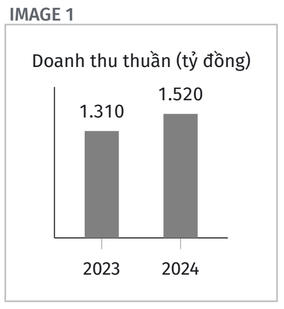
A base **Qwen3-VL-4B** transcribes every chart, **26 LoRA-tuned Qwen3.5-4B** writers each write one program, and a **plurality vote** over their canonical programs picks the answer. Every model has at most 8B parameters.

1 Combining writers that *differ* helps more than decoding one writer more often.
2 A single writer's errors start early, are shared across writers, and resisted every training remedy we tried.

1 Task: ViTNumChart

Answer a numerical question over a **Vietnamese financial report** (paragraphs, HTML tables, chart images) by writing a **reasoning program**. Arithmetic steps (`add`, `subtract`, `multiply`, `divide`, `greater`) take numbers or references `#k`; retrieval steps `table_op`, `chart_op` copy labels from the document.

IMAGE 1



INPUT

Question (translated)
By how much did net revenue change from 2023 to 2024?

Context: report paragraphs, HTML tables, up to 13 chart images

OUTPUT: A PROGRAM, SCORED BY PROGRAM ACCURACY (PA)

gold `chart_at(Image 1; Doanh thu thuần; 2024; none); chart_at(Image 1; Doanh thu thuần; 2023; none); subtract(#0; #1)`

error `chart_at(Image 1; Doanh thu; 2024; none); …`
label not as printed on the chart

error `subtract(1520; 1310)`
right value, but typed from the bars

2 Data and official results

	Train	Public	Private
Questions	1725	459	980
Documents	74	19	39
Distinct images	187	61	83
Chart-step gold	678	202	–

The public test is development data only and is never used for training.

Submission	Public	Private	
	strict	PA	EA
1 Vote, 26 writers	74.07	**0.7917**	0.8916
2 + views, 32 writers*	74.51	**0.7917**	**0.8933**
3 Larger vote	74.51	0.79	–
4 HTML-table writer	73.42	0.7711	–

*Leaderboard entry; private values as printed by the platform. EA (value) exceeds PA (program): many errors lie in the program's *form*.

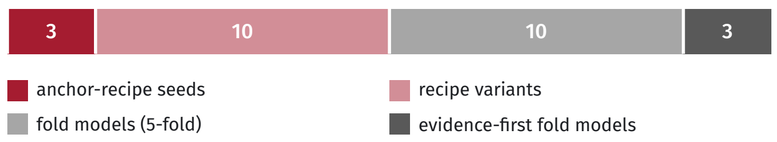
Where the vote helps (public strict %, descriptive)

Slice	*n*	Anchor	Vote
Arithmetic only	182	76.4	**81.3**
Chart step	202	67.8	**70.3**
Unseen structure	43	11.6	**20.9**

3 Training the writers

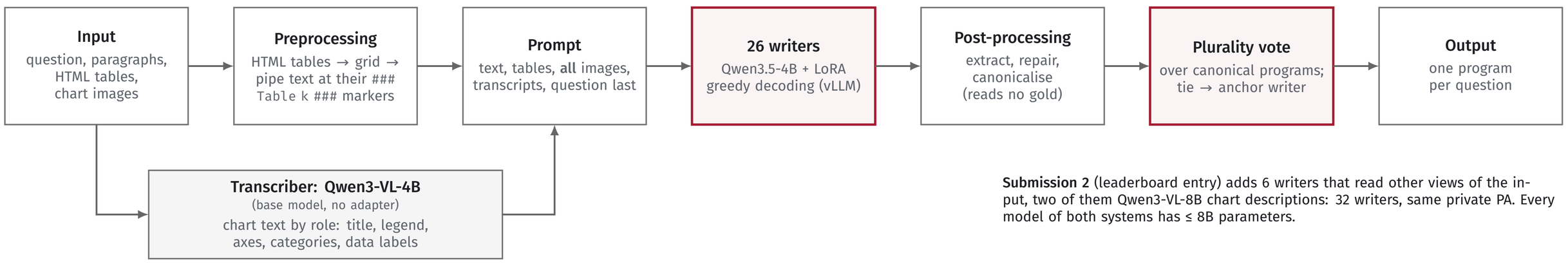
- Each training question becomes one chat example: the inference prompt, with the gold program as target.
- **LoRA** r=32, α=64 on attention, MLP and Gated DeltaNet projections; vision tower frozen.
- lr 10^{-4}, 3 epochs, bf16, prompts ≤ 6,144 tokens.
- Median 0.87 h per writer on one RTX 5090.
- Training prompts attach the gold images plus 1–2 distractor images; inference attaches *every* image.
- Five document-level folds for cross-fitting; no external data.

The 26 vote members (submission 1)



All share backbone and prompt; they differ in target format, training folds, seed and objective, so that their *errors* differ.

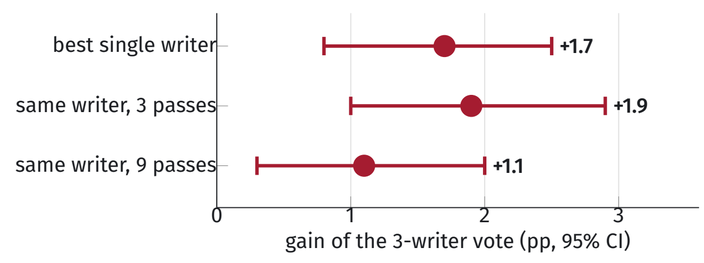
4 System

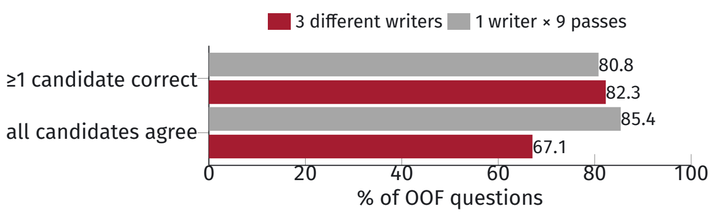


Submission 2 (leaderboard entry) adds 6 writers that read other views of the input, two of them Qwen3-VL-8B chart descriptions: 32 writers, same private PA. Every model of both systems has ≤ 8B parameters.

5 Diverse writers vs. more samples

Out of fold (1725 questions), a vote of **three different writers** vs. one writer voting over its own samples:

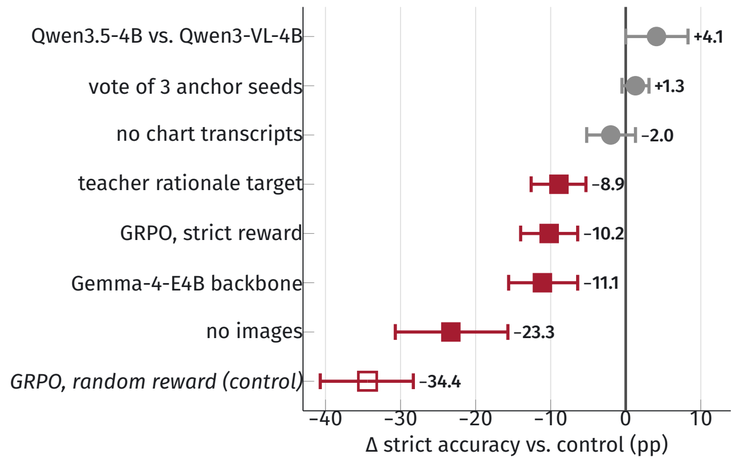




Different writers agree less, but hold a correct program more often.

6 Ablations and training remedies

Public set, wild cluster bootstrap-t, Holm over 23 tests.
■ survives Holm ● not resolved

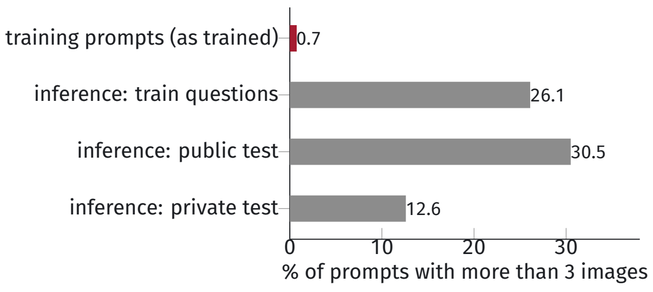


Zero-shot Qwen3-VL-4B scores 12.6%, so fine-tuning is needed. Of 13 remedies (target, objective, capacity, data) none helped and two hurt; minimum detectable effect ≈ 4.95 pp.

7 Images help; extra images hurt

- Removing all images: **−23.3 pp** (public set).
- Blanking chart images on 678 OOF chart questions: **−13.4 pp**.
- Blanking images the gold program *does not read*: **+1.72 pp** [0.74, 2.73] on 1047 OOF questions.

A likely cause is a train/inference mismatch:



Next step: train with the inference-time image policy.

8 Where a single writer loses accuracy

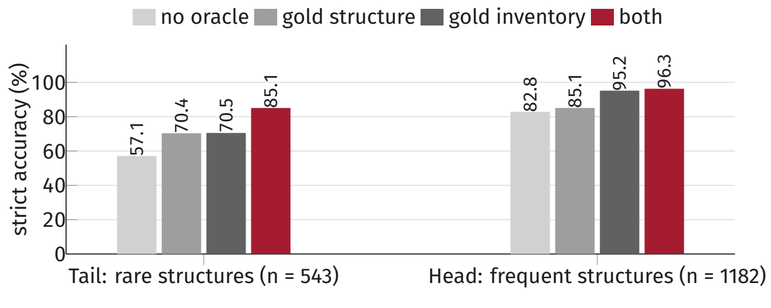
83.8%
of wrong multi-step programs already leave the gold path **at step 0**
38.5% if every step failed independently

17.7%
of OOF questions are failed by **all three** writers
5.1% expected if errors were independent

88.1% vs. 76.8%
a correct program is in the candidate pool, vs. chosen by the **best gold-free selector**

9 Structure and inventory oracles

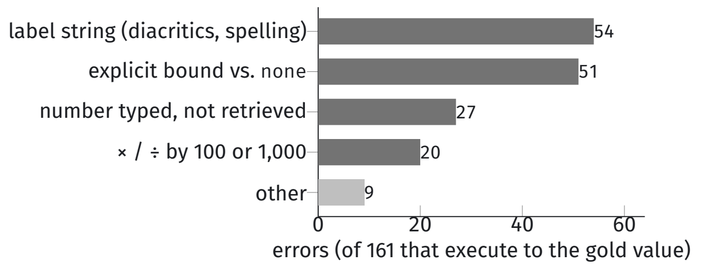
A pre-specified factorial on 1725 OOF questions gives the writer the gold *structure* (operators and #k dataflow), the gold *inventory* (argument strings), or both.



Inventory +12.70 pp, structure +5.74 pp; both together reach 92.75%. A chance-matched control cut the inventory's lead from +8.4 to +2.8 pp.

10 Right value, wrong form

161 of 435 strict OOF errors (37%) execute to the gold value. Many reflect the Vietnamese presentation of the data: diacritic labels and *triệu* / *tỷ* scales.



Picking the gold form among a writer's own samples would reach 77.62% (from 74.78%), yet a convention model trained to make that choice changed accuracy by −0.17 pp.

11 Conclusion

- 26 LoRA-tuned 4B writers and a 4B transcriber ranked **first** under the 8B limit.
- Combining **writers that differ** beat more passes of one writer.
- A single writer's errors **begin early**, are **shared**, and resisted every training remedy.
- Code, adapters and data were shared with the organisers; a vote replay reproduces the submissions exactly.

References

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